28.12.2022

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# TPE-001

## B. TECH. (FIFTH SEMESTER)

## END SEMESTER EXAMINATION, Dec., 2022

ANTENNA AND WAVE PROPAGATION

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each question carries 10 marks.

1. (a) Define the terms antenna efficiency, antenna gain, antenna directivity, antenna bandwidth and antenna beamwidth. (CO1)

   (b) Determine the HPBW and FNBW of the given electric field : (CO1)

   $$\vec{E} = 12 \cos\theta \sin 2\phi \, e^{-jkr}$$

   (c) Explain the reciprocity theorem and duality theorem with proper examples. (CO1)

2. (a) Define the term polarization. By writing the general expression of electric field provide the conditions for linear, circular and elliptical polarization. Determine the state of polarization for given electric field :

   $$\vec{E} = \left(2\hat{a}_\theta + 3j\hat{a}_\phi\right) e^{jkr}$$

*P. T. O.*

If circular or elliptical, also determine the sense of rotation. In determination of state of polarization and sense of rotation, write each step clearly. (CO2)

(b) Show that the radiation resistance of small circular loop antenna is $20\pi^2 \left(\frac{C}{\lambda}\right)^4$, if the electric field of small loop antenna is given as : (CO2)

$$E_\phi = \frac{\eta k^2 a^2 I_0 e^{-jkr}}{4r} \sin\theta$$

where symbols have their usual meaning.

(c) Discuss the helical antenna in normal and axial mode. (CO2)

3. (a) Show that the Array Factor (AF) of four elements broadside linear array having non-uniform amplitude and equal spacing between the consecutive elements is given as : (CO3, CO6)

$$AF = a_1 \cos\left(\frac{kd}{2}\cos\theta\right) + a_2 \cos\left(\frac{3kd}{2}\cos\theta\right)$$

where symbols have their usual meaning.

(b) Show that the array factor of N-element array having uniform amplitude, progressive phase shift and equal spacing between the consecutive elements is given as : (CO3, CO6)

$$AF = \sin\left(\frac{N\psi}{2}\right) / N \sin\left(\frac{\psi}{2}\right)$$

where symbols have their usual meaning.

(c) Write the short notes on Yagi-Uda antenna and Log-perodic antenna. (CO3, CO6)

4. (a) Explain various feeding techniques applied to parabolic reflector antennas. Explain in detail Cassegrain and Gregorian feeding system. (CO4, CO6)

(b) Discuss the different feeding techniques for patch antenna with proper diagram. (CO4, CO6)

(c) Design a rectangular microstrip antenna with dimension W and L, over a single substrate, whose center frequency is 10 GHz. The dielectric constant of the substrate is 10.2 and the height of the substrate is 0.127 cm. Determine the physical dimensions W and L of the patch taking account field fringing. (CO4, CO6)

5. (a) Explain the tern critical frequency, maximum usable frequency, skip distance and virtual height. Establish the relationship between critical frequency and maximum usable frequency. (CO5)

(b) Explain sub-refraction, super refraction and duct propagation in reference to tropospheric propagation. (CO5)

(c) A radio communication link is to be established via the ionosphere. The virtual height at the midpoint of the path is 300 km and the critical frequency is 9 MHz. Determine the maximum usable frequency for the link between the stations of distance 800 km assuming the flat earth condition: (CO5)

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# TEC–501

## B. TECH. (FIFTH SEMESTER)

## END SEMESTER EXAMINATION, Dec., 2022

DIGITAL SIGNAL PROCESSING

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks.

1. (a) Find the inverse Z-transform of the following using Long Division method : (CO1)

$$X(z) = \frac{1}{\left(1 - 1.5z^{-1} + 0.5z^{-2}\right)}$$

when :

(i) ROC $|z| > 1$

(ii) ROC $|z| < 0.5$

(b) Determine the DTFT and Energy Density Spectrum of the sequence : (CO1)

$$x(n) = \begin{cases} A; & 0 \le n \le L - 1 \\ 0; & \text{Otherwise} \end{cases}$$

*P. T. O.*

(c) Check whether the following digital systems and linear, time invariant and causal or not : (CO1)

$$y(n) = x^2(n)$$

2. (a) State and proof the "circular time shifting" property and "Parseval's relation" of DFT. (CO2)

(b) Prove that the multiplication of two DFTs is equivalent to circular convolution. (CO2)

(c) Compute N = 8 point DFT of given below sequence using DITFFT algorithm : (CO2)

$$x(n) = 2^n;\, 0 \le n \le 7$$

3. (a) Prove the Direct Form II structure of the IIR digital filter. Draw its Signal flow graph and its Transposed Structure. (CO3)

(b) Determine the cascade and parallel realization of the following systems : (CO3)

$$H(z) = \frac{3\left(4 + 5z + 2z^2\right)}{(2z+1)(z+2)}$$

(c) Prove the draw the structure for linear phase FIR Filter for M even. Determine and draw the linear phase realization of FIR, whose system function is : (CO3)

$$H(z) = \left(\frac{2}{3z^{-1}} + \frac{2}{3z^{-2}}\right)$$

4. (a) Prove the 'Bilinear Transformation Technique' for design of IIR filter. (CO4)

(b) Prove the "Impulse Invariant" transformation techniques for IIR digital filter. Convert the analog bandpass filter with system function

$$H(s) = \frac{1}{(s + 0.1)^2 + 9}$$

into a digital IIR filter by use of the backward difference for the derivatives. (CO4)

(c) Explain the design steps of Chebyshev filter. (CO4)

5. (a) Explain Hamming, Hanning and Blackman, Rectangle, Bartlett and Kaiser windows for design of FIR filter. (CO5, CO6)

(b) The desired response of low-pass filter is :

$$H_d\left(e^{j\omega}\right) = e^{-j2\omega}; \quad \frac{-\pi}{4} < \omega < \frac{\pi}{4}$$

$$= 0; \quad \frac{\pi}{4} < \omega \le \pi$$

Determine $H\left(e^{j\omega}\right)$ if window function is defined as : (CO5, CO6)

$$w(n) = \begin{cases} 1, & 0 \le n \le n \\ 0, & \text{otherwise} \end{cases}$$

(c) What is "Gibbs phenomenon" ? Why is this phenomenon developed in rectangular window and it is eliminated or reduced ? Discuss the windowing method of FIR filter design. (CO5, CO6)



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# TEC–502

## B. TECH. (ECE) (FIFTH SEMESTER)

## END SEMESTER EXAMINATION, Dec., 2022

COMMUNICATION SYSTEMS—II

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) For a sinusoidal modulating signal $m(t) = A\cos 2\pi f_m t$, prove that the maximum output signal to quantization noise ratio in a DM system under the assumption of no slope overload is given by :

$$(\mathrm{SNR})_o = \left(\frac{S}{N_q}\right)_0 = \frac{3f_s^3}{8\pi^2 f_m^2 f_M}$$

where $f_s = \frac{1}{T_S}$ is the sampling rate and $f_M$ is the cutoff frequency of a low pass filter at the output of the receiver. (CO1)

*P. T. O.*

(b) (i) A signal $x_1(t)$ is band limited to 2 kHz while $x_2(t)$ is band limited to 3 kHz. Find the Nyquist rate for (I) $x_1(2t)$ (II) $x_2(t-3)$ (III) $x_1(t)+x_2(t)$.

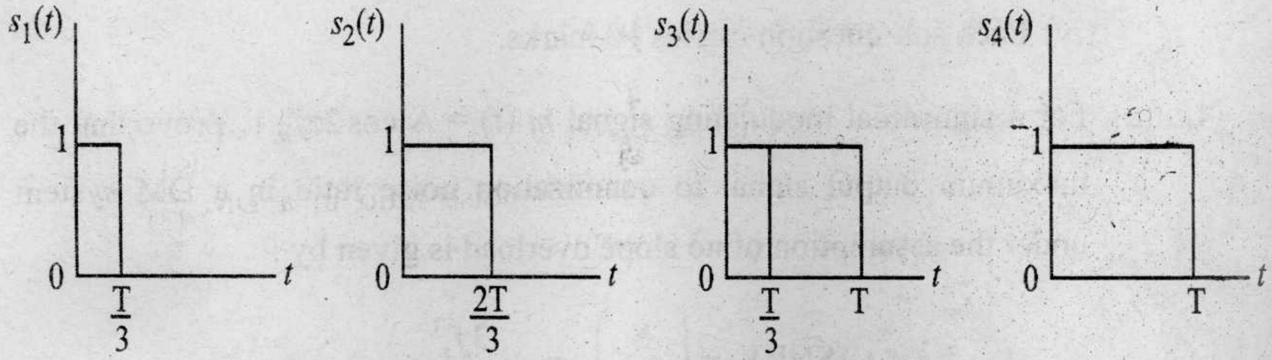
(ii) Three analog signals, having bandwidths 1200 Hz, 600 Hz and 600 Hz, are sampled at their respective Nyquist rates, encoded with 12 bit words, and time division multiplexed. Determine the bit rate for the multiplexed signal. (CO1)

(c) What is the slope overload distortion and granular noise in delta modulation and how it removed in adaptive delta modulation ? (CO1)

2. (a) Determine the power spectral density for the unipolar NRZ format. (CO2)

(b) Give detailed proof of Gram-Schmidt orthogonalization procedure. (CO2)

(c) Figure shown below displays the waveforms of four signals $s_1(t), s_2(t)$, $s_3(t)$ and $s_4(t)$. Using the Gram-Schmidt orthogonalization procedure, find an orthonormal basis for this set of signals. (CO2)




3. (a) Prove that the impulse response of the matched filter is a time reversed and delayed version of the input signal. (CO3)

(b) Write a short note on raised cosine spectrum. (CO3)

(c) Explain the Nyquist criterion for distortionless baseband binary data transmission. (CO3)

4. (a) Draw the block diagram of QPSK system and explain its working. (CO4)

(b) Derive an expression for the spectrum of BFSK and sketch it for the orthogonality and minimum bandwidth. (CO4)

(c) Derive an expression for probability of error of BPSK wave. (CO4)

5. (a) Prove that the upper bound on the value of entropy H (X) of a source X is $\log_2 M$ where M is the size of the alphabet of X. (CO5 & CO6)

(b) A DMS X has four symbols $x_1, x_2, x_3$ and $x_4$ with $P(x_1) = 1/2$, $P(x_2) = 1/4$ and $P(x_3) = P(x_4) = 1/8$. (CO5 & CO6)

(i) Construct as Shannon-Fano code for X, and calculate the efficiency of the code.

(ii) Repeat for the Huffman code and compare the results.

(c) Write the advantages and disadvantages of a digital communication system. (CO5 & CO6)

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# TEC–503

## B. TECH. (FIFTH SEMESTER)

## END SEMESTER EXAMINATION, Dec., 2022

## MICROCONTROLLER AND EMBEDDED SYSTEMS

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each sub-question carries 10 marks.

1. (a) With neat internal block schematic, explain the architecture of 8051. (CO1)

(b) The following shows crystal frequency for three different 8051-based systems. Find the period of the machine cycle in each case. (CO1)

(i) 11.0592 MHz

(ii) 16 MHz

(iii) 20 MHz

(iv) 22 MHz

(c) Explain each PORT available in 8051. (CO1)

*P. T. O.*

2. (a) Write a program to add two 32 bit numbers with and without carry. (CO2)

   (b) Explain the addressing modes available in 8051 microcontroller with examples to each. (CO2)

   (c) Write a program to transfer value FEH serially by pin P2.1. Put two highs at the start and end of the data. Send the byte LSB first. (CO2)

3. (a) Write a program to generate two square waves one of 5 kHz frequency at pin P1.3 and another of frequency 25 kHz at pin P2.3. Assume XTAL = 22 MHz. (CO3)

   (b) Assuming XTAL = 22 MHz, write a program to generate a square wave of ON time 3 ms and a OFF time of 10 ms on all pins of Port 0. (CO3)

   (c) Design a counter for counting the pulses of an input signal. The pulses to be counted are fed to pin P3.4. XTAL = 22 MHz. (CO3)

4. (a) Explain the important features of ATMEGA328P. Also explain the utility of Ports. (CO4)

   (b) Explain the sketch of Arduino Software and basic analog commands used. (CO4)

   (c) Write a program where according to the push button value at pin 7 the LED at Pin 13 works by using Arduino and also display the value of LED serially. (CO4)

5. (a) Explain the concept of interfacing of ADC 0804 chip with timing diagram. (CO5, CO6)

(b) Write a program to interface LCD for sending command and data.

(CO5, CO6)

(c) Two switches are connected to pins P3.2 and P3.3. When a switch is pressed, the corresponding lines goes low, write a program to :

(CO5, CO6)

(i) light all LEDs connected to port 0, if the first switch is pressed.

(ii) light all LEDs connected to port 2, if the second switch is pressed.



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# TEC–504

## B. TECH. (ECE) (FIFTH SEMESTER)
## END SEMESTER EXAMINATION, Dec., 2022

CONTROL SYSTEM

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

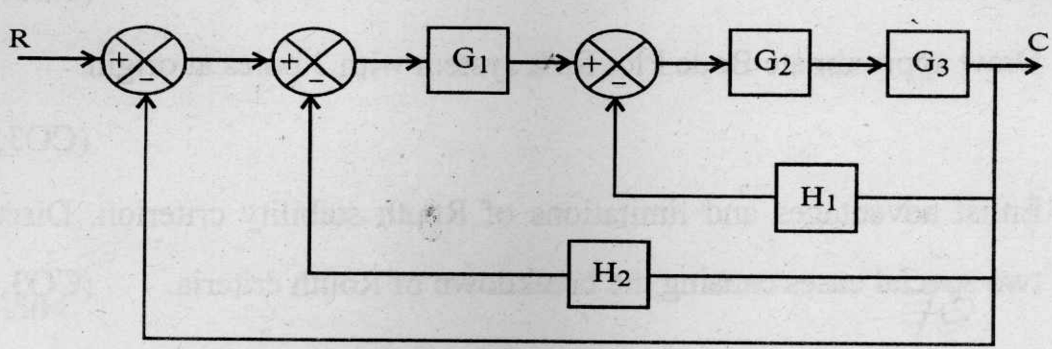
(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Derive an expression for the closed loop transfer function of a feedback control system. (CO1)

(b) What is a source node, sink node, dummy node and forward path gain transmittance in case of a signal flow graph ? Demonstrate with help of an example. Explain the Mason's gain formula with each and every term used in this formula. (CO1)

(c) Determine the overall gain of the block diagram using reduction technique for the given block diagram of the system. (CO1)




*P. T. O.*

2. (a) Derive the expressions for Peak time and Peak overshoot for a second order system. (CO2)

   (b) Derive a Unit step response of a first order closed loop control system. (CO2)

   (c) Analyze the effect of incorporating PD controller to the closed loop second order control system. (CO2)

3. (a) Discuss various steps to draw a Polar Plot for given open loop transfer function. (CO3)

   (b) For a unity feedback control system : (CO3)

   $$G(s) = K/\{s(s+1)(s+2)\};$$

   Sketch Root Locus showing all important details on this plot.

   (c) Consider the characteristic equation : (CO3)

   $$s^4 + 2s^3 + (4 + K)s^2 + 9s + 25 = 0$$

   Using the Routh's stability criterion, determine the range of K for stability.

4. (a) Draw an electrical RC phase lag network and derive its transfer function. (CO3, 4)

   (b) Draw approximate Bode Plot for a system with 3 poles at origin. (CO3, 4)

   (c) Enlist advantages and limitations of Routh stability criterion. Discuss two special cases causing the breakdown of Routh criteria. (CO3, 4)

5. (a) Derive homogenous solution of state space equation. (CO5, 6)

(b) What are the limitations of transfer function model and advantages of state space model ? (CO5, 6)

(c) Derive state space model for a system represented by third order differential equation given by : (CO5, 6)

$$d^3y(t)/dt^3 + 4\,d^2y(t)/dt^2 + 7\,dy(t)/dt + 8\,y(t) = 9\,u(t).$$

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# TEC–551

## B. TECH. (ECE) (FIFTH SEMESTER)

## END SEMESTER EXAMINATION, Dec., 2022

SENSOR TECHNOLOGY

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the following terms : (CO1)

   (i) Transfer function

   (ii) Reliability

   (iii) Uncertainty

   (b) Explain Hysteresis error in detail with diagram. (CO1)

   (c) Write short notes on the following : (CO1)

   (i) Full scale output

   (ii) Accuracy

2. (a) Explain about different capacitance level sensor in detail. (CO2, CO3)

*P. T. O.*

(b) Explain the follow terms in detail with proper diagram : (CO2, CO3)

(i) Theromoelectric

(ii) Pyroelectric effect

(c) Explain Halt-effect, and describe its application in detail. (CO2, CO3)

3. (a) Describe the need of interface electronic circuits. (CO3)

(b) Explain noise in sensors and its classification. (CO3)

(c) Describe light to voltage and voltage to light converters. (CO3)

4. (a) Describe in detail about potentiometric sensor. (CO4)

(b) Explain about optoelectronic motion detectors. (CO4)

(c) Explain capacitive occupancy sensor with proper diagram. (CO4)

5. (a) Describe capacitive and resistive humidity sensors. (CO5)

(b) What are image sensors ? Explain CMOS sensor in detail. (CO5)

(c) Explain gas flame detectors. (CO6)

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# TEC–591

## B. TECH. (ECE) IoT (FIFTH SEMESTER)

## END SEMESTER EXAMINATION, Dec., 2022

TRANSDUCERS, ACTUATORS AND DISPLAY DEVICES

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is the difference between Piezoresistive and Piezoelectric effect ? Explain with example. (CO1)

   (b) A radiation of wavelength 250 mm is incident on a surface having 4.8 eV work function. Will photo electron be observed ? (CO1)

   (c) Describe various static characteristics of sensors. (CO1)

2. (a) A common base photo transistor has input impedance of 15 ohm and output resistance of 100 kilo ohm. If a signal of 250 mv is applied between emitter and base. Find the voltage amplification. Assume alpha = 1. (CO2)

(b) Explain working of Interferometric optical transducer. (CO2)

(c) Explain any *one* solid state transducer in detail. (CO2)

3. (a) Explain piezoelectric transducer. How is it different from Electro-strictive transducer ? (CO3)

   (b) What is microsensing for MEMS ? Differentiate between capacitive sensing and piezoelectric sensing. (CO3)

   (c) What is the working principle of Electromechanical transducers ? (CO3)

4. (a) Explain PN homo junction solar cell with diagram and its I-V characteristics. (CO4)

   (b) What is the efficiency and spectral response of a solar cell ? (CO4)

   (c) What is the effect of temperature and light intensity on solar cell properties ? (CO4)

5. (a) What is Plasma Display ? Explain electroluminescence with diagram. (CO5, CO6)

   (b) What is the difference between LED and LCD. (CO5, CO6)

   (c) Find out the output power generated by a solar cell of 4 $m^2$ area. If the input radiation is 850 $W/m^2$ and the efficiency of solar cell is 25%. (CO5, CO6)